OpenClaw Proactive Interaction

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1. Course Content

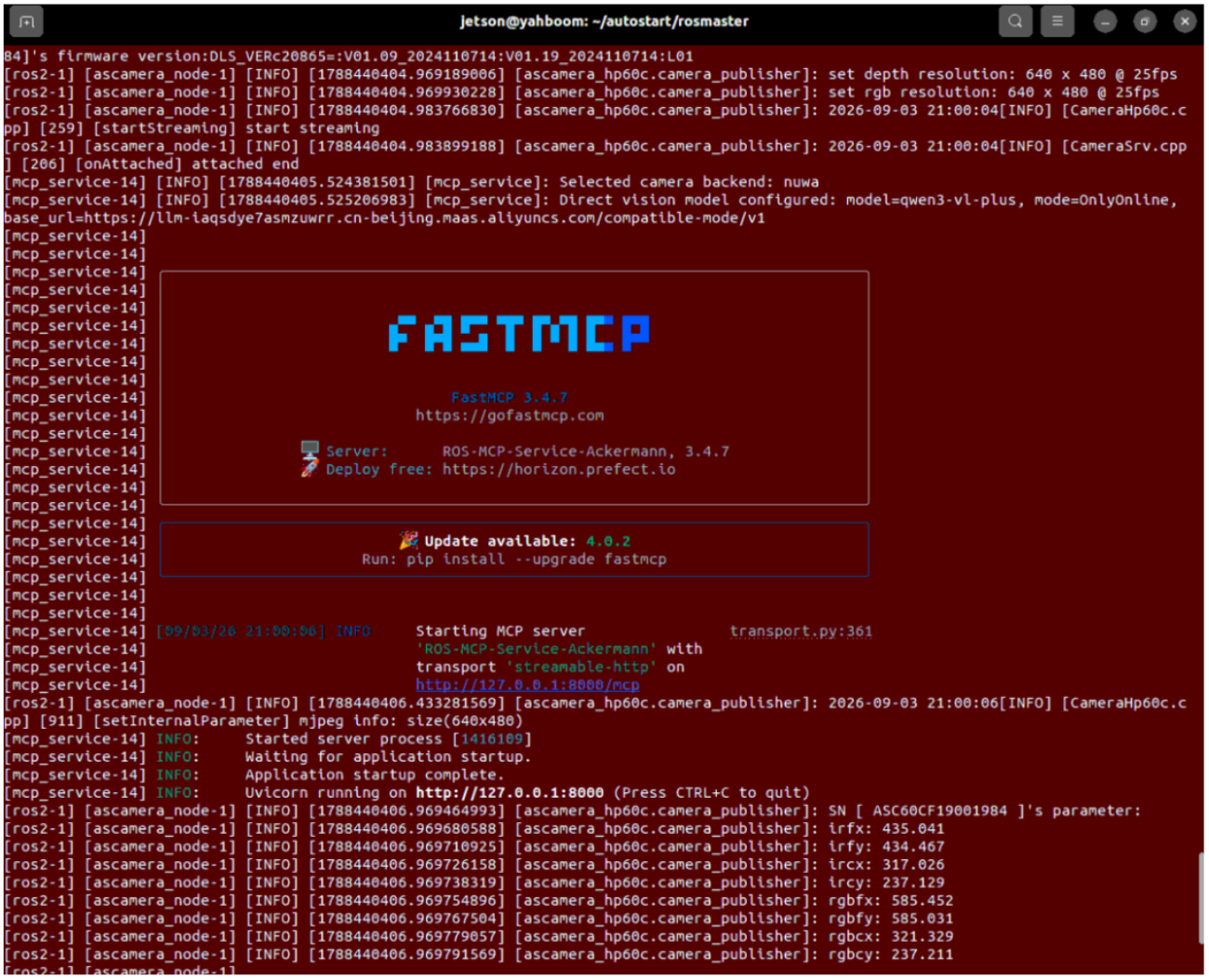
Course Overview

- Based on skills, write response strategies for specific scenarios, enabling OpenClaw to proactively provide solutions when facing unexpected situations
- This section is a scenario-oriented solution design course. It is recommended to complete the prerequisite courses 【OpenClaw Basics】【OpenClaw Interaction Methods】【OpenClaw Skills Development】first

2. Preparation

- First study the 【OpenClaw Skills Development】 chapter to master creating new skills

```
ros2 launch dofbot_pro_bringup mcp_service.launch.py
```



3. Create a New Skill

- Refer to 【OpenClaw Skills Development】 to create a response strategy skill for special situations, and enable the skill

  Note: If you encounter garbled text when copying and pasting on VNC, this is due to the remote desktop issue. It is recommended to use VSCode SSH remote login for copying and pasting

```
---
name: material-shortage-process
description: Provides a preset handling workflow for material shortages encountered during inspection tasks.
version: 1.0.1
---

## Production Workshop Color Material Shortage Handling Plan
- Proactively offer solutions to the user via inquiry. The corresponding strategy must only be executed after user confirmation.
- For example, if the quantity of a specific color material is observed to be less than 2, it is classified as a material shortage.
## Option 1: Replenish from the raw material warehouse shelves
## Option 2: Go to the storage area and replenish using materials of the same specifications but a different color
```

- After creation, restart the OpenClaw gateway to take effect

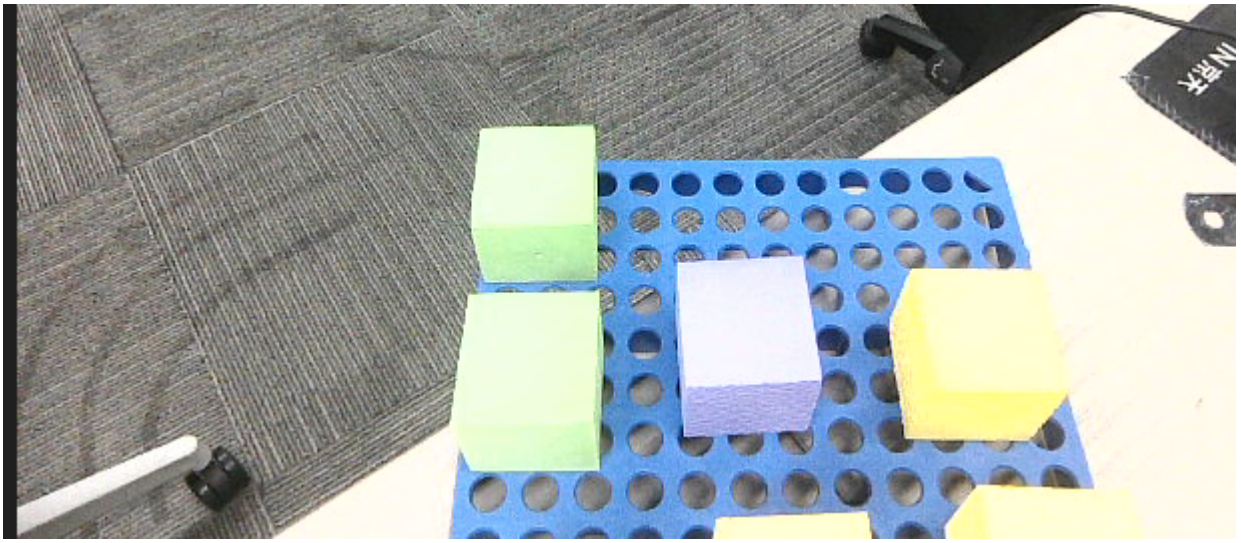
```
openclaw gateway restart
```

4. Case Demonstration

> **Note**
>
> - Commands can be written according to your own needs. The following is a case demonstration. You can choose any interaction method. Here, the web-based WebChat is used as an example

- Here, OpenClaw will proactively provide handling suggestions based on actual observed situations combined with skill requirements. The same situation does not require users to issue commands repeatedly each time





5. Extended Applications

OpenClaw's proactive interaction capability can be applied to various scenarios. The following provides several reference custom plans:

5.1 Equipment Fault Early Warning and Handling

```
---
name: equipment-anomaly-process
description: Proactively provides diagnostic and handling suggestions when abnormal
equipment operating parameters are detected.
version: 1.0.0
---

## Proactive Equipment Anomaly Warning Plan
- Monitor equipment operating status in real-time and proactively alert the user when an
anomaly is detected.
- Provide tiered handling suggestions: prompt for minor anomalies, recommend shutdown for
severe anomalies.
- The corresponding handling strategy must only be executed after user confirmation.

## Option 1: Parameter Calibration
- Check if the sensor readings are accurate.
- Execute the equipment self-test program.
- Calibrate the deviation parameters.

## Option 2: Shutdown for Maintenance
- Stop the equipment immediately.
- Notify maintenance personnel to be on-site.
- Record fault logs for subsequent analysis.
```

5.2 Production Progress Tracking and Reminders

```
---
name: production-progress-reminder
description: Periodically checks the progress of production tasks and proactively reminds
the user at key milestones.
version: 1.0.0
---

## Proactive Production Progress Tracking Plan
- Periodically query the completion status of production line tasks.
- Proactively remind the user at key milestones (e.g., material shortage, schedule delay,
task completion).
- Provide suggestions for next steps.

## Reminder Strategy
- Material running low: Remind to replenish 2 hours in advance.
- Progress falling behind schedule: Analyze the reasons and provide suggestions to catch up.
- Task successfully completed: Generate a report and notify relevant personnel.
```

5.3 Multi-Condition Compound Trigger Strategy

```
---
name: smart-condition-response
description: Provides intelligent proactive responses by integrating multi-dimensional
conditions such as time, environment, and equipment status.
version: 1.0.0
---

## Multi-dimensional Condition Response Plan
- Comprehensively analyze various sensor data and business logic.
- Proactively trigger a response when a specific combination of conditions is met.
- Provide personalized handling suggestions.

## Trigger Condition Examples
- Time condition: During the night shift (22:00-06:00).
- Environmental condition: Temperature exceeds 35°C and humidity is below 30%.
- Equipment condition: Continuous operation for more than 8 hours.

## Response Strategy
- Proactively recommend equipment rest and maintenance.
- Remind staff to pay attention to environmental safety.
- Automatically adjust equipment operating parameters to energy-saving mode.
```